



Andrei Silviu Dragnea

Senior Rust Backend Engineer / Contractor / Freelancer

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SKILLS

C	9 years
Operating Systems	8 years
Rust	5 years
Java	5 years
Kotlin	5 years
Spring	5 years
Spring Boot	5 years
Maven	5 years
Scala	5 years
Mockito	5 years
JUnit	5 years
TestNG	5 years
Docker	5 years
Async frameworks	5 years
Functional programming	5 years
MySQL	3 years
Cassandra	3 years
Redis	3 years
Memcached	3 years
Kafka	2 years
Kubernetes	2 years
Grafana	2 years
Prometheus	2 years

EDUCATION

Faculty of Automatic Control and Computers,
Politehnica University of Bucharest
Bachelor in Computer Science
2013-2017

ABOUT ME

I am a Senior Rust Backend Engineer / Contractor / Freelancer with **6+ years of experience**, only interested in remote work.

Contributor to **Spring Framework** with bug fixes and feature improvements.

Contributor to **Docker** with documentation improvements.

Ex-Adobe engineer in **Identity Management** System and **Video Analytics** teams.

Do you need software solutions for your business? Drop me a message!

MOST RELEVANT WORK EXPERIENCE

Senior Rust Software Engineer | Adobe Rust web application

May 2021 - Present

The client, Adobe Video Analytics, part of the Adobe Experience Cloud, had significant infrastructure costs on the video events collection system because of it being a JVM application.

- **Reduced CPU usage by 2.6x and memory usage by 51x** by rewriting the API server from Scala to Rust.

Skills: Rust, Axum, Actix Web, Warp, Scala, Serde, Tokio, Futures, Prost, Moka, Cargo, RdKafka, Async, Kubernetes, Helm, Kapitan

Senior Scala Software Engineer | Adobe Traffic filtering

May 2021 - Present

The client, Adobe Video Analytics, part of the Adobe Experience Cloud, had significant infrastructure costs because of traffic generated by one important customer which did not pay a subscription.

- **Reduced infrastructure costs by 35% (\$1.4M per year)** by implementing a traffic filtering mechanism in the video event collection system. The ID of every video session has an embedded boolean flag which indicates if the events of that session are dropped or not.

Skills: Scala, Akka, Kafka, Spark, cats-effect, Caffeine, Circe, Maven, Protobuf, sbt, Kubernetes, Maven, Docker, Helm, Kapitan, Jenkins

Senior Scala Software Engineer | Adobe Service migration from Java 8 to Java 17

May 2021 - Present

The client, Adobe Video Analytics, had a legacy application that needed new features and developers could not be productive using an old Java version.

- **Updated legacy application from Java 8 to Java 17**, by first bringing all the dependencies of the application to the latest versions, then bumping the Java version.

Skills: Java, JDK, Mockito, Spring, Spring Boot, JUnit, TestNG, Apache Tomcat, Docker

- Senior Scala Software Engineer |  Adobe

System resilience improvement

May 2021 - Present

The client, Adobe Video Analytics, part of the Adobe Experience Cloud, had an application with unpredictable performance on traffic spikes or when an upstream service became unavailable.

 - **Improved service resilience by 12x** by correctly handling blocking code in the context of a non-blocking application, thus reducing on-call incident rates.
 - **Improved service fault-tolerance by 15x** by isolating the blocking writes to each of our upstreams to its own blocking thread, thus limiting the impact of a faulty upstream to only a fraction of customer traffic.

Skills: Scala, Java, Kafka, cats-effect, ExecutorService, CompletableFuture, Concurrency
- Senior Scala Software Engineer |  Adobe

Automatic dependency updates

May 2021 - Present

The client, Adobe Experience Cloud, needed periodic updates to the dependencies of all code repositories of the organization, both for security reasons and new features.

 - **Reduced operational costs related to dependency updates by 95%** by installing a bot which automatically applies and validates dependency version updates.
 - **Contributed to the Renovate open-source project** with bug fixes for dependency updates in our application.

Skills: GitHub, Renovate, Jenkins, Docker, Earthly
- Senior Scala Software Engineer |  Adobe

Build time optimization

May 2021 - Present

The client, Adobe Video Analytics, part of the Adobe Experience Cloud, had an application with very slow compile times, which reduced developer productivity significantly.

 - **Reduced project compilation time by 3x, from 3 min to less than 1 min** by optimizing JSON codec generation at compile time and migrating the Scala project build system from Maven to sbt.

Skills: Scala, Circe, IntelliJ IDEA, Maven, sbt
- Senior Scala Software Engineer |  Adobe

Jenkins Job DSL

May 2021 - Present

The client, Adobe Experience Cloud, needed a way to automatically bootstrap a new CI server from a version-controlled specification.

 - **Reduced Jenkins Pipeline jobs creation time by 95%** by using Jenkins Job DSL. I also added static analysis, test results and coverage reports for both successful and failing CI builds, in order to improve developer productivity.

Skills: Jenkins, Jenkins Pipeline DSL, Jenkins Job DSL, Jenkins Code Coverage API, Jenkins Warnings Next Generation Plugin
- Senior Scala Software Engineer |  Adobe

Duplication removal

May 2021 - Present

The client, Adobe Video Analytics, part of the Adobe Experience Cloud, had many components in its systems which used duplicated code, increasing the risk of bugs when updating one copy and leaving another behind.

 - **Reduced code duplication by 80%** by extracting a module containing common code used by multiple components of our application, thus also reducing coupling between modules.

Skills: Scala, Refactoring, Scala Kubernetes Client, Circe, Maven, Kafka, Async Programming, cats-effect, ScalaMock, ScalaTest

- **Senior Java Software Engineer |  Adobe**
Azure Active Directory Synchronization server May 2018 - May 2021
The client, Microsoft Azure Active Directory, wanted to automate the management of identity data between Microsoft and Adobe systems.
 - Reduced identity data import time **by 99%, from a few days to 15 minutes**, by implementing a SCIM2 server for pushing user and group creation and update events from Microsoft Azure AD to Adobe Identity System.**Skills:** Java, Kotlin, Kotlin coroutines, Spring, Spring Boot, Project Reactor, Netty, Redis, Mockito, JUnit, TestNG, SCIM2, Refactoring, Maven, Gradle, Docker, Jenkins, Azure Active Directory, Azure Graph API, Jakarta EE
- **Senior Java Software Engineer |  Adobe**
Re-use the same business logic for both a blocking and non-blocking application May 2018 - May 2021
The client, Adobe Identity Management System, wanted to reduce infrastructure costs by scaling only the traffic-heavy components of a monolithic application under heavy load.
 - Reduced infrastructure costs **by 70%** by extracting a core business logic component used both in the legacy monolithic blocking application and in a new non-blocking microservice. The decoupled component was re-written from Java to Kotlin using coroutines.**Skills:** Java, Kotlin, Kotlin coroutines, Refactoring, IntelliJ, Gradle, Maven, Spring, Spring Boot, MySQL, Cassandra, WildFly, Java EE
- **Senior Java Software Engineer |  Adobe**
Simplified system design May 2018 - May 2021
The client, Adobe Identity Management System, wanted to reduce infrastructure costs by simplifying the traffic flow for the most used identity APIs.
 - Reduced traffic through internal systems **by 60% (from 100K RPM to 40K RPM)** and instance count **by 70% (from 30 to 9)** by bypassing an intermediate service. I extracted a common business logic module called both by the core service and intermediate service. Then I made the core service call the target service directly, instead of the intermediate one.**Skills:** Java, Spring, Maven, Refactoring
- **Senior Java Software Engineer |  Adobe**
Created clean tests framework May 2018 - May 2021
The client, Adobe Identity Management System, wanted to help its developers write more maintainable tests.
 - Improved developer productivity **by 50%** by creating a clean tests framework on top of Mockito, TestNG, JUnit 4, JUnit 5 and Spring TestContext framework. This framework automatically guides developers into writing decoupled tests with minimum setup.**Skills:** Java, Mockito, Spring, Spring Boot, SpringTestContext, Junit 4, JUnit 5, TestNG



Senior C Software Engineer | Instacar

Car sharing application

September 2017 - December 2017

The client, a car sharing company, wanted to allow customers to control and monitor rented cars remotely through a mobile application.

- **Reduced operational costs by 90%** by implementing a TCP server over a proprietary binary protocol for remote car control. The cars could be unlocked and have their gas level checked automatically from a monitoring dashboard.
- **Improved application resilience by 80%** by recompiling the Linux kernel with custom TCP settings in order to compensate for a faulty TCP implementation on the devices used for remote car control.
- **Reduced codebase size by 30%** by writing a PyCharm plugin for automatic dead code elimination. The codebase also had to be annotated with Python 3 type hints in order to guide the dead code elimination process.

Skills: C, Linux kernel, TCP, Binary Protocols, PyCharm, IntelliJ



OTHER EXPERIENCE:

Open source contributor to:

- Spring Framework
- Docker
- Mockito
- Project Reactor
- Scala Kubernetes Client
- Renovate
- Reactor Netty
- IntelliJ Kotlin
- IntelliJ Community

Creator of open-source **Automatic Recursive Git Rebaser**

IntelliJ plugin developer: **Pycharm Python Enhancements**